Monotonic Array

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The problem

- Is the array entirely non-increasing or entirely non-decreasing?

Intuition

- An array is monotonic if it never goes both up and down.

Approach

- 1. Track two flags, non-increasing and non-decreasing
- 2. Clear the opposite flag on each rise or fall
- 3. Return true if either flag survives

Java solution

```java
public class MonotonicArray {
    public static boolean isMonotonic(int[] a) {
        boolean inc = true, dec = true;
        for (int i = 1; i < a.length; i++) {
            if (a[i] < a[i - 1]) inc = false;
            if (a[i] > a[i - 1]) dec = false;
        }
        return inc || dec;
    }
}
```

Complexity

- Time: O(n)
- Space: O(1)

Recap

- An array is monotonic if it never goes both up and down.
- Complexity: O(n)